

Deep Learning Detected Nutrient Deficiency in Chili Plant

Arief Rais Bahtiar
Magister Informatika
Universitas Atma Jaya Yogyakarta
Yogyakarta, Indonesia
ariefraisb@gmail.com

Albertus Joko Santoso
Magister Informatika
Universitas Atma Jaya Yogyakarta
Yogyakarta, Indonesia
albjoko@staff.uajy.ac.id

Pranowo
Magister Informatika
Universitas Atma Jaya Yogyakarta
Yogyakarta, Indonesia
pranowo@uajy.ac.id

Jujuk Juhariah
Faculty of Agriculture
Universitas Boyolali
Boyolali, Indonesia
jujukjuhariah@gmail.com

Abstract— Chili is a staple commodity that also affects the Indonesian economy due to high market demand. Proven in June 2019, chili is a contributor to Indonesia's inflation of 0.20% from 0.55%. One factor is crop failure due to malnutrition. In this study, the aim is to explore Deep Learning Technology in agriculture to help farmers be able to diagnose their plants, so that their plants are not malnourished. Using the RCNN algorithm as the architecture of this system. Use 270 datasets in 4 categories. The dataset used is primary data with chili samples in Boyolali Regency, Indonesia. The chili we use are curly chili. The results of this study are computers that can recognize nutrient deficiencies in chili plants based on image input received with the greatest testing accuracy of 82.61% and has the best mAP value of 15.57%.

Keywords— deep learning, chili plant, object detection, nutrient, region convolutional neural network

I. INTRODUCTION

Human experts diagnose plant deficiency as basically subjective and limited to the area and supporting infrastructure [1]. Plant nutrients are divided into two, namely, macronutrients and micronutrients [2]. The computer vision algorithm can change this problem with faster prediction results with the convolutional neural networks (R-CNN) region [3]. This success brought computer vision resolution with the R-CNN model for a variety of classification and detection tasks that were faster than the convolutional neural networks (CNN) model. CNN is considered to be still slow to detect large and complex amounts of data [4]. When the CNN model becomes a model for complex object segmentation, the process is considered ineffective because the model will take all the proposed areas in each image. The R-CNN Mask Model is the solution to this problem by taking the proposal regions to be detected [5]. The R-CNN mask uses the basic R-CNN Faster extractor to recognize objects in the mask.

Computer vision is mostly done in the field of pattern recognition and agriculture. One of them is authenticating the multi-style text of the Qur'an [6]. The result can obtain an effective accuracy of 87.1%. One branch of computer vision is image processing, which is used to diagnose human and plant diseases [7]. This study uses 876 samples, and the accuracy is more than 90%. The text extraction algorithm is also used to recognize text in text scenes and image documents and can increase the efficiency of the OCR process by 15% - 20% [8]. Besides, computer vision is also applied in other matters relating to disease detection. Research on alfalfa leaf disease by

researchers using the SVM model resulted in training and testing accuracy of 97.64% and 94.74 [9]. The result from 129 features is only 45 of the best features for this SVM model. Similar identification is also used to analyze and measure soybean leaf damage as a guide to applying insecticides with image processing models [10]. The results show the quantification of leaf damage with a precision comparable to expert science. To handle deeper data, the convolution neural network (CNN) model is the solution [11]. The application of CNN to identify diseases has been carried out to detect 13 types of leaf diseases [12]. The Caffe framework used in this study produced a precision of between 91% and 98% for the model developed while for separate class tests, an average of 96.3%. CNN was also able to identify four types of apple leaf disease with an accuracy of 97.62% [13]. This study uses a total of 1053 images and is assisted by experts to classify the type of disease. CNN is also used in detecting cassava disease [14]. This study uses 720 images, and videos can reduce f1-score by 32%.

Here, we investigate the diagnosis of nutrient deficiencies in 3 network architectures. We tested the performance of the R-CNN object detection model for diagnoses of nutrient deficiency in chili plants in agriculture. In each nutrient deficiency category, we tested four levels of macronutrient deficiency symptoms, to assess the performance of the model for early detection of symptoms. We report accuracy, memory, F-1 scores, and accuracy for an image to assess R-CNN performance.

II. METHOD

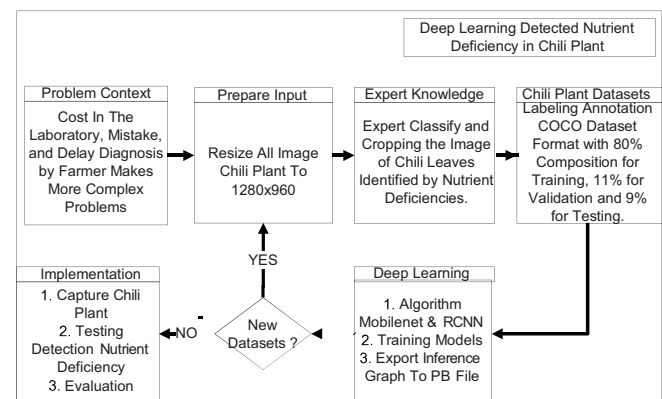


Fig. 1. Flow Method

We use the Tensorflow platform to use the R-CNN object detection model designed to identify leaf symptoms from three types of nutrient deficiencies and healthy leaves

in chili (*Capsicum annum L*). We use the Single Shot Multibox (SSD) model with MobileNet detection and classification and Mask RCNN Inception V2 with Faster R-CNN detection and classification with COCO (Common Objects in Context) dataset. For simplicity, we refer to the CNN object detector model as the cellular CNN model. We use transfer learning to perfect the model parameters to our dataset consisting of 270 chili leaf images for four classes. The chili leaf dataset is made with pictures taken at the Argo Ayuningtani Farmers Group in Boyolali District, Indonesia. Full details of this dataset can be seen in Table 1.

TABLE I. EXPERT VALIDATED DATASET RESULTS

Leaf Indications	Chili
Iron Deficiency	17
Magnesium Deficiency	27
Mangan Deficiency	8
Nitrogen Deficiency	100
Phosphorus deficiency	3
Potassium Deficiency	131
Healthy	71
Sulfur Deficiency	19
Calcium Deficiency	57
Zinc Deficiency	0
Total	433

For this study, chili experts made a classification and agreed on annotations. Initially, the expert received the primary data from the results of preparing the input for the classification of types of nutrient deficiencies. The nutrient elements used in this study are macronutrients. According to experts, macronutrients are considered easier to detect visually if compared to macronutrients. Fig 2 shows the types of macronutrients used in this study are calcium deficiency, nitrogen deficiency, potassium deficiency, and healthy leaves as a compliment. At this stage, experts will subjectively classify their diagnosis of chili leaves that are indicated to lack macronutrients and healthy leaves. The findings are marked by cropping images of chili plants. This study uses diagnosis 1 figure 1 type of nutrient deficiencies to facilitate labeling in the process of making chili plant datasets. Similar studies have been carried out to detect cassava using the CNN method based on expert judgment [14].



Fig. 2. Examples of Detected Plants

There are three different annotation styles tested to identify class objects: (1) mask all leaves that have dominant symptoms, (2) mask a portion of the leaflet around the core symptoms, and (3) a combination of annotation styles (1) and (2) combined with the same class label for all leaflets and inside the leaflet mask. Based on

training results of up to 5000 epochs on the 48GB NVIDIA GeForce GTX 1080, the second annotation style recorded the lowest overall loss and was chosen to test on a desktop device in the field.

We chose four classes for field detection - nitrogen (ND) deficiency, potassium deficiency (PD), calcium deficiency (CD), and healthy leaves (H). For simplicity, ND, PD, CD, and H are referred to collectively as nutrients in the next text. This nutrient was chosen because it is an obstacle affecting chili production in Indonesia [15]. Making chili plant datasets based on the results of the chili plant expert classification the total dataset is 270 images.

A. Data Preprocessing

The chili leaf dataset from JPEG images was taken with a Nikon Coolpix digital camera. Full details of this dataset were previously reported in Table I. For this study; chili plant experts extracted 270 images from the dataset based on the visibility of the most severe symptoms of each class. At this stage, the researchers made observations to obtain primary data on chili plants. This data is in the form of a picture of a chili leaf. This study took a sample of chili plants in Boyolali Regency, Indonesia. In addition to collaborating with the local government, this study also collaborated with the Argo Ayuningtani Farmer Group, Senden Village, Selo District, Boyolali Regency, Indonesia, as a place for taking curly chili samples. Sampling began in August 2019. Primary data which can then be resized to 1280x960 before being given to experts for classification. The dataset consists of 4 classes, namely ND (71), PD (71), H (71), and CD (57). Then the validated dataset is divided into three parts, i.e., 80% training data, 11% validation data, and 9% testing data per category. The results can be seen in Table II. The next step is to make labeling of the data in the form of COCO Format. After the 3-part labeling process, JSON files are obtained for each part. This JSON file is used to create TFRecord files by using a library from the Tensorflow API. The TFrecord is then implemented in the Tensorflow environment for the training process. This TFRecord file is a combination of all images and annotations compressed in one file. Each model is trained 5000 epochs.

TABLE II. RESULTS OF SELECTED DATASET DISTRIBUTION

Category	Chili Dataset (<i>Capsicum annum L</i>)		
	Training	Validation	Testing
ND	57	8	6
PD	57	8	6
H	57	8	6
CD	45	7	5
Total	216	31	23

At the time of the field, it was also known that the busy schedule of farmers starting from farming in the morning until noon than in the afternoon going to the market to sell the harvest made the supporting factor of the emergence of nutrient deficiencies in the chili plants. So, the time to report or consult agricultural problems to extension agents or agricultural experts is difficult, and farmers tend to use instant subscriptions with chemical fertilizers without knowing what nutrient detection plants need.

B. RCNN Models

We evaluate the performance of RCNN models that are built using standard precision metrics and independent evaluations based on automatic and manual data matching. R-CNN is a development of CNN for image recognition that focuses on an object, often called a support vector machine (SVM). R-CNN is implemented in many fields one of them, R-CNN, is applied in the field of detection of high-resolution remote sensing objects [16]. R-CNN is also used to detect faces to find out the benchmark value with the dataset [17]. The R-CNN process can be seen in Fig 3.



Fig. 3. Remote Sensing Object Detection Process on R-CNN [16]

For the object detection model architecture, we chose the Single ShotMultibox (SSD) model and RCNN Inception V2 Mask with MobileNet and Faster R-CNN detection and classifier [18] [5]. This model is used because it is one of the fastest object detection models available through Tensorflow [19]. The SSD model performs the task of localizing objects and classifying objects in one forward motion on a mobile device [20]. While the RCNN Mask model performs the task of localizing objects and classifying objects in the proposal region from the input data. Pre-trained RCNN SSD and Mask models trained on the COCO (Common Objects in Context) dataset downloaded from Tensorflow's Detection Model Zoo [19] and transfer learning is used to perfect the model parameters. COCO is a large-scale object detection, segmentation, and captioning dataset consisting of 330 K images, 1.5 million object instances, and 80 object classes. Each model is trained up to 500 epochs using 15 batch sizes on 2 NVIDIA Tesla V100GPU on Azure, Microsoft cloud computing, and storage platforms. Hyperparameter SSD Mobilenet V2 and SSDLite Mobilenet V2 models were selected as follows: initial learning rate 0.004, iou threshold 0.6, batch size 24, while the RCNN Inception V2 Mask model initial learning rate 0.0002, iou threshold 0.7, batch size 1. This model trained 5000 epochs. We will compare the mAP of the three models to find out which evaluation models are best for our dataset.

III. RESULT

The training results for the three types of models produce total loss, as in Table III. The training results show that the R-CNN inception v2 mask model produces the lowest total loss of 0.0588 and takes 17 minutes. These results show the COCO dataset format that we created matches the RCNN Inception v2 mask model.

TABLE III. TRAINING RESULTS

SSD Mobilenet V2		SSDLITE Mobilenet V2		Mask RCNN Inception V2	
Total Loss	Time (min.)	Total Loss	Time (min.)	Total Loss	Time (min.)
2.4124	43	1.8347	41	0.0588	17

After training, the results of inference from each model were tested with 23 test drawings or 9% of the prepared dataset. The purpose of this test is to find the best accuracy model from the chili plants dataset, as shown in Table IV. The table contains total True Computer Detection (T), Incorrect Computer Detection (F), and Computer Cannot Detect (N) per topology based on expert validation data. For simplicity, T, F, and N are referred to as test results in the next text collectively:

TABLE IV. TEST RESULT

Detection Result	Models		
	SSD Mobilenet V2	Mask RCNN Inception V2	SSDLITE Mobilenet V2
$\sum T$	5	19	1
$\sum F$	7	4	0
$\sum N$	11	0	22
Testing Accuracy	21.74%	82.61%	4.35%

Fig 4 is the result of a comparison of testing images with healthy leaf detection results. The result is the computer can detect healthy leaves in the image. Whereas in Fig 5 shows the results of the detection of calcium nutrient deficiency.



Fig. 4. Testing Results 1



Fig. 5. Testing Results 2

Table 5 shows the mAP details of the three models used. The result is the Mobilenet V2 SSD model has the largest mAP value of 15.57%, and the RCNN Inception V2 Mask has the fastest time.

TABLE V. DETAILS MAP SCORE

SSD Mobilenet V2		SSDLITE Mobilenet V2		Mask RCNN Inception V2	
mAP (%)	Time (min.)	mAP (%)	Time (min.)	mAP (%)	Time (min.)
0.1557	19	0.05826	20	0.146	14

IV. FINDING RESEARCH

There are some difficulties in detecting the types of nutrient deficiencies in chili leaves based on the results of testing or implementation in the field. The emergence of multi detection in one leaf as in Fig 6. Detection means that the plant is deficient in calcium, but in the detection appears detection of calcium, healthy and nitrogen. This proves that the RCNN Mask method has not been able to classify in as much detail as desired by experts. Do not rule out the leaves are already complex, so the computer detects multi detection. But this can still be handled by giving instructions to conclude the results of detection to farmers on the application to be built. If there is a case like the above, the leaves lack calcium. Because the area is bigger than nitrogen deficiency. If this is a healthy leaf, it is certainly not possible because there are indications of two nutrient deficiencies.

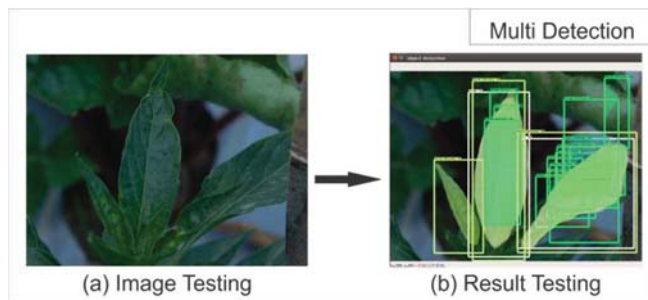


Fig. 6. Problem Detection

V. CONTRIBUTION

The contribution of this research is to help farmers to detect nutrient deficiencies in their chili plants by using deep learning. Deep learning is based on expert knowledge. Aside from being a detection medium for nutrient deficiencies, deep learning nutrient deficiency also compares the use of 3 RCNN topologies to the chili plant dataset. Exploring deep learning technology in agriculture specifically detecting nutrients so as not to be malnourished and making chili plant datasets is a contribution of this research.

VI. CONCLUSION

Failure to harvest chili plants occurs one of them due to farmer mistakes in detecting plant nutrients. This is due to the lack of initiative by farmers to look for information and report problems to extension workers or experts. The reason for distance and time is the obstacle. This is because farmers are busy planting to selling to the vegetable market. Deep learning technology can recognize the lack of nutrients from chili plants based on imagery with an accuracy of 82.61% and the best mAP value of 15.57%. The precision and timing of the image will affect the detection process. In the future, the concept of RCNN Mask can be carried out further research in the form of mobile apps, so farmers can find out in real-time and more quickly be able to identify the types of nutrient deficiencies in their chili plants.

VII. ACKNOWLEDGMENTS

We would like to thank the Boyolali Regency Government, the Boyolali Regency Agriculture Office, the

Selo District Agricultural Extension Center and the Argo Ayuningtani Farmers Group for their assistance and permission to extract the chili plant primary data.

REFERENCES

- [1] C. H. Bock, G. H. Poole, P. E. Parker, and T. R. Gottwald, "Plant Disease Severity Estimated Visually, by Digital Photography and Image Analysis, and by Hyperspectral Imaging," *CRC. Crit. Rev. Plant Sci.*, vol. 29, no. 2, pp. 59–107, 2010.
- [2] Q. Chen *et al.*, "Autophagy and Nutrients Management in Plants," *Cells*, vol. 8, no. 11, pp. 1–17, 2019.
- [3] M. Kang, K. Ji, X. Leng, and Z. Lin, "Contextual Region- Based Convolutional Neural Network with Multilayer Fusion for SAR Ship Detection," *Remote Sens*, vol. 9, no. 8, p. 860., 2017.
- [4] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into High Quality Object Detection," in *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 6154–6162.
- [5] K. He, G. Gkioxari, P. Dollar, and R. Girshick, "Mask R- CNN," in *The IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2961–2969.
- [6] S. Hakak, A. Kamsin, S. Palaiahnakote, and O. Tayan, "Residual-based approach for authenticating pattern of multi-style diacritical Arabic texts," pp. 4–6, 2018.
- [7] N. Petrellis, "A Review of Image Processing Techniques Common in Human and Plant Disease Diagnosis," *Symmetry (Basel)*, vol. 10, no. 270, pp. 2–35, 2018.
- [8] P. Sahare and S. B. Dhok, "Review of Text Extraction Algorithms for Scene- text and Document Images," *IETE Tech. Rev.*, vol. 34, no. 2, pp. 144–164, 2017.
- [9] F. Qin, D. Liu, B. Sun, L. Ruan, Z. Ma, and H. Wang, "Identification of Alfalfa Leaf Diseases Using Image Recognition Technology," *PLoS One*, vol. 11, no. 12, pp. 1–26, 2016.
- [10] B. Brandoli *et al.*, "BioLeaf: A professional mobile application to measure foliar damage caused by insect herbivory," *Comput. Electron. Agric.*, vol. 129, pp. 44–55, 2016.
- [11] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep Convolutional Neural Networks for Hyperspectral Image Classification," *J. Sensors*, vol. 2015, 2015.
- [12] S. Sladojevic, M. Arsenovic, A. Anderla, D. Culibrk, and D. Stefanovic, "Deep Neural Networks Based Recognition of Plant Diseases by Leaf Image Classification," *Comput. Intell. Neurosci.*, vol. 2016, 2016.
- [13] B. Liu, Z. Yun, D. He, and L. Yuxiang, "Identification of Apple Leaf Diseases Based on Deep Convolutional Neural Networks," *Symmetry (Basel)*, vol. 10, no. 1, pp. 1–16, 2018.
- [14] A. Ramcharan *et al.*, "A Mobile-Based Deep Learning Model for Cassava Disease Diagnosis," *Front. Plant Sci.*, vol. 10, p. 272, 2019.
- [15] Hapsah, Gusmawartati, A. I. Amri, and A. Diansyah, "Respons Pertumbuhan dan Produksi Tanaman Cabai Keriting (*Capsicum annum* L .) terhadap Aplikasi Pupuk Kompos dan Pupuk Anorganik di Polibag," *J. Hortik. Indones.*, vol. 8, no. April, pp. 203–208, 2017.
- [16] Y. Cao, X. Niu, and Y. Dou, "Region-based Convolutional Neural Networks for Object Detection in Very High Resolution Remote Sensing Images," *2016 12th Int. Conf. Nat. Comput. Fuzzy Syst. Knowl. Discov.*, pp. 548–554, 2016.
- [17] H. Jiang and E. Learned-miller, "Face Detection with the Faster R-CNN," in *2017 12th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2017)*, 2017, pp. 650–657.
- [18] W. Liu *et al.*, "SSD: Single Shot MultiBox Detector," *Lect. Notes Comput. Sci.*, pp. 21–37, 2016.
- [19] Google, "Tensorflow Detection Model Zoo.," 26-Jan-2017.
- [20] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using Deep Learning for Image-Based Plant Disease Detection," *Front. Plant Sci.*, vol. 7, p. 1419, 2016.